A Mechanistic Study of AI-Detection Neurons in Frozen BERT: Sparse Probing and Activation Patching on RAID

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Abstract

AI-generated text detectors achieve high accuracy on standard benchmarks, yet the internal representations that drive these predictions remain poorly understood. We study which neurons in a *frozen*, unmodified BERT-base-uncased encoder support AI-text detection, using the RAID benchmark across six generators spanning pure-base and instruction-tuned architectures. Applying the L1→L2 sparse-probing protocol of Gurnee et al. (2023) to all 9,216 CLS hidden-state dimensions (12 layers $\times$ 768), which we term *neurons* throughout, we identify stable sets of 45–62 neurons per generator ($<$1% of neurons); a full-feature L2 probe achieves 90.5–98.8% detection accuracy, while a probe restricted to the stable set retains 86.5–97.2%, both with high cross-fold reproducibility (Jaccard 0.67–0.76). To assess the causal role of these neurons, we apply same-domain bidirectional activation patching: transplanting AI activations into human targets (forward) flips 1.07–8.15% of predictions to AI, 10–16$\times$ above a random-neuron baseline; transplanting human activations into AI targets (reverse) flips 0.65–5.74%, 7–12$\times$ above random, confirming partial necessity. Mean-ablating the same set leaves accuracy largely intact ($\leq$1.1 pp drop on 5/6 generators), suggesting the signal is redundantly distributed. Cross-generator analysis reveals a bipartite structure: instruction-tuned generators concentrate 30–36% of stable neurons in BERT's final layer, while both base generators fall below 14%, consistent with a layer-12 footprint of post-training alignment. Leave-one-family-out evaluation shows that the selected neurons retain 86–94% of the full-feature ceiling on entirely unseen generator families.

1 Introduction

The rapid development of Large Language Models (LLMs) has made AI-generated text increasingly difficult to distinguish from human writing (Dugan et al., 2024). This poses a significant challenge across domains ranging from academic integrity to social media moderation and journalist authenticity. Although detection accuracy has improved substantially, with encoder-based classifiers reaching high performance on established benchmarks (Dugan et al., 2024), a fundamental question remains unanswered: *What internal representations drive these predictions?* Mechanistic interpretability (Elhage et al., 2021; Bereska and Gavves, 2024) has seen little application to AI-text detection, leaving the field without a causal explanation of how detection works.

Existing interpretability work on AI-text detection takes several forms: (a) *black-box feature attribution* via SHAP (Lundberg and Lee, 2017) or LIME (Ribeiro et al., 2016), which identifies predictive tokens without inspecting model internals (Pudasaini et al., 2026); (b) *confounder removal*, which suppresses neurons correlated with domain or style artefacts in fine-tuned encoders to improve out-of-distribution accuracy (Borile and Abrate, 2025); (c) *sparse autoencoder feature analysis* applied to decoder-based models (Kuznetsov et al., 2025); (d) *probing on fine-tuned detectors*, which reveals reliance on narrow lexical or structural cues (Choi and Zu, 2025); and (e) *topological analysis* of attention maps, characterising surface and syntactic properties that differentiate AI from human text (Kushnareva et al., 2021). These approaches share a fundamental limitation documented across neuron-level analysis broadly (Sajjad et al., 2022): they are *observational*. They identify neurons that correlate with detector behaviour or remove neurons that hurt generalisation, but none directly tests whether a small set of internal representations is *causally sufficient* to produce AI-text predictions.

We address this gap with a causal intervention study on a *frozen* BERT-base-uncased; Figure 1 summarises the experimental pipeline. Rather than

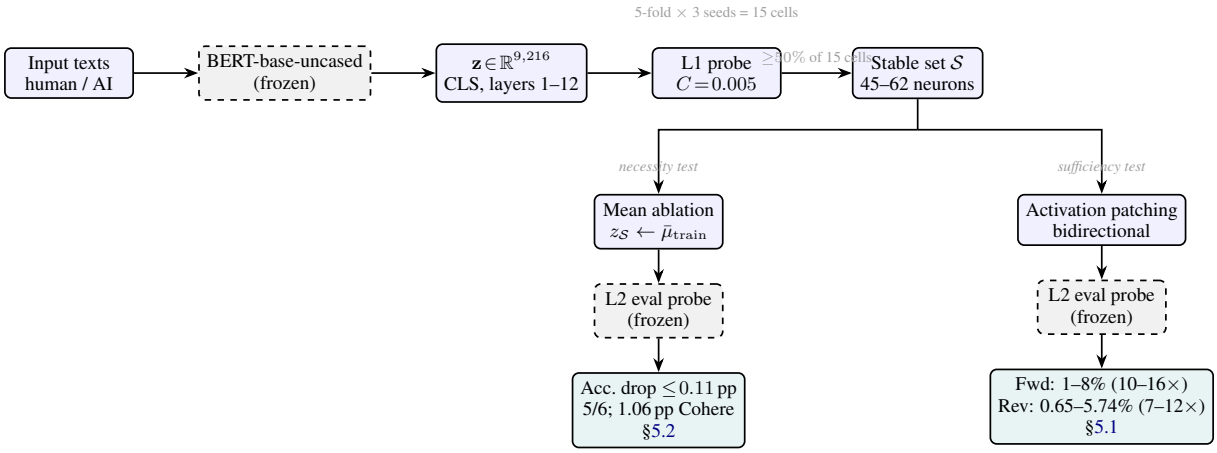

Figure 1: Overview of the experimental pipeline. Frozen BERT-base-uncased CLS activations are concatenated across all 12 layers, yielding $\mathbf{z} \in \mathbb{R}^{9,216}$. An L1-regularised probe ($C = 0.005$) selects a sparse candidate set per (fold, seed) cell; neurons stable in $\geq 50\%$ of all 15 cells become the *stable set* $\mathcal{S}$ (45–62 neurons per generator). An independent L2 probe trained on all 9,216 neurons is frozen and reused as the decision function in both experiments. **Left branch** (§5.2): mean ablation (≤ 0.11 pp drop on 5/6 generators; 1.06 pp Cohere). **Right branch** (§5.1): bidirectional activation patching: forward (AI $\rightarrow$ human) flips 1–8% of human samples to AI, 10–16$\times$ above random; reverse (human $\rightarrow$ AI) flips 0.65–5.74% of AI samples to human, 7–12$\times$ above random.

fine-tuning the model for detection, we train sparse linear probes on pretrained CLS-token activations concatenated across all 12 layers (9,216 neurons total), following the sparse-probing protocol of Gurnee et al. (2023). An L1-regularised logistic regression selects a small stable set of candidate neurons; an independent L2 probe serves as the frozen evaluator for all subsequent experiments. We then apply two causal interventions to the same selected neuron set: *mean ablation*, which replaces these neurons with their training-set means to test necessity; and *same-domain bidirectional activation patching*, run in both the forward direction (AI donor $\rightarrow$ human target, testing sufficiency) and the reverse direction (human donor $\rightarrow$ AI target, testing partial necessity on-manifold). The intervention orientation is the key distinction from prior work: rather than suppressing suspected confounders, we inject donor activations and measure whether predictions flip.

Across six generators spanning two pure-base and four instruction-tuned families, bidirectional activation patching establishes that the selected neurons are both causally sufficient (forward patching flips human predictions to AI at 10–16$\times$ above a random-feature baseline) and partially necessary (reverse patching flips AI predictions to human at 7–12$\times$ above random), while mean-ablation of the same neurons leaves accuracy largely intact. The detection signal is therefore *redundantly distributed*: easy to inject, hard to erase. Cross-generator analysis reveals generator-specific stable sets that nonetheless overlap $\sim$24$\times$ above chance, with a layer-12 concentration in instruction-tuned generators not present in either base model. Leave-one-family-out evaluation confirms that the selected neurons retain most of the full-feature detection signal on entirely unseen generator families.

Our work is most directly related to Borile and Abrate (2025), who suppress confounding neurons in a fine-tuned encoder to improve out-of-distribution detection, an observational approach that identifies which neurons hurt generalisation rather than which neurons causally drive predictions. More broadly, Antverg and Belinkov (2022) show that individual-neuron analysis in language models is prone to false positives without causal validation (a pitfall also observed in recent AI-text-detection work (Enkhbayar, 2025)). We address this gap directly via activation patching, and discuss the relationship to this and other prior work in Section 2.

Contributions. We make four contributions: (i) a sparse-probing characterisation of AI-detection signal in frozen BERT-base-uncased across six RAID generators, identifying stable sets of $<1\%$ of CLS dimensions sufficient for near-ceiling detection accuracy with high cross-fold stability; (ii) the first application of bidirectional activation patching to AI-text detection neurons, establishing causal sufficiency in the forward direction (10–16$\times$ above

random) and partial necessity in the reverse (7–12× above random); (iii) a cross-generator circuit analysis that reveals generator-specific stable sets which nonetheless overlap ∼24× above chance, with a layer-12 footprint concentrated in instruction-tuned generators and absent in both pure-base models; and (iv) a leave-one-family-out (LOGO) evaluation showing that the selected neurons retain 86–94% of the full-feature detection signal on entirely unseen generator families.

2 Related Work

2.1 AI-Generated Text Detection

Supervised fine-tuned classifiers (Solaiman et al., 2019; Zellers et al., 2019) achieve high in-distribution accuracy but degrade substantially on unseen generators or domains. Zero-shot methods such as DetectGPT (Mitchell et al., 2023) and Binoculars (Hans et al., 2024) exploit log-probability statistics without labelled data but require whitebox or API access to a reference model. Systematic benchmarking via RAID (Dugan et al., 2024), spanning 11 generators and 6 domains, confirms that no single method generalises reliably across generators or domains. This motivates our focus on what internal representations *drive* detection, rather than improving detection accuracy itself.

2.2 Interpretability of Text Detectors

The shallowest form of interpretability for AI-text detectors uses input-level attribution. Pudasaini et al. (2026) apply SHAP to explain classifier decisions, finding that the most predictive tokens diverge substantially across domains. This approach characterises the model's decision surface but cannot speak to what internal representations underlie it.

More recent work examines representations inside the encoder. Borile and Abrate (2025) identify neurons in a fine-tuned BERT encoder whose activations correlate with domain or style artefacts rather than AI-authorship signal, and suppress them to improve out-of-distribution accuracy. Kuznetsov et al. (2024) take a geometric approach, removing a learned embedding subspace to produce a detector more robust to lexical variation. Both works operate on fine-tuned models and test necessity-for-confounding: the question is which neurons hurt generalisation, not which neurons are sufficient to drive predictions. Our work differs from Borile and Abrate (2025) on four axes: we use a *frozen* rather than fine-tuned encoder; we probe CLS-token representations rather than layer-specific FFN sublayer activations; we use ablation to measure *necessity* and additionally apply activation patching to test causal *sufficiency*, rather than ablating to identify and suppress confounders; and we evaluate on RAID rather than DAIGT, M4 (Wang et al., 2024), or HC3 (Guo et al., 2023).

Kuznetsov et al. (2025) apply sparse autoencoders to Gemma-2 decoder activations, extracting semantically interpretable features correlated with AI authorship. This work is complementary but distinct. The model class (autoregressive decoder), representation type (token-position activations rather than CLS pooling), and analysis method (SAE reconstruction) all differ from ours. The approach also remains observational: features are identified by correlation, not by causal intervention. None of the above provides direct evidence that a small identified set of neurons is *sufficient* to produce AI-text predictions when injected into human-text activations; that is the question we address.

2.3 Sparse Probing and Activation Patching

Earlier work identified task-relevant neurons in pretrained models via correlation-based selection, and mapped salient neurons to linguistic properties across layers and model families (Durrani et al., 2023). Sparse probing uses L1-regularised linear classifiers to identify a small set of neurons in a pretrained model that carry task-relevant information. Gurnee et al. (2023) introduced and systematically evaluated the L1-then-L2 protocol: an L1 probe selects candidate neurons, and a L2 probe re-fitted on those neurons provides unbiased accuracy estimates. Their study shows that individual neurons in GPT-2 and LLaMA encode grammatical, factual, and linguistic properties, and that the selected sets are stable across seeds and data subsets. We adopt this protocol directly, applying it to BERT-base-uncased CLS-token activations concatenated across all 12 layers.

Activation patching replaces internal representations in a target forward pass with those from a donor pass, then measures the effect on model outputs (Vig et al., 2020; Meng et al., 2022). Heimersheim and Nanda (2024) recommend donor patching with real inputs (rather than zero or mean vectors) to avoid out-of-distribution artefacts, and mean ablation as a conservative necessity baseline;

we follow both. The closest methodological precedent for our flip-rate framing is Ravindran (2025) (donor patching, flip-rate metric; autoregressive decoder, safety alignment). Activation patching has also been applied to BERT cross-encoders for IR (Lu et al., 2025), but to our knowledge no prior work applies it to encoder-based AI-text detection.

3 Setup

3.1 Data: RAID

RAID (Dugan et al., 2024) provides human-written and machine-generated texts across 11 generators and 8 domains. We use six of the eight RAID domains: abstracts, books, news, Reddit posts, reviews, and Wikipedia. We exclude poetry and recipes, as both have very low sample counts in RAID and risk introducing genre-specific artefacts that confound cross-domain analysis. We select six generators spanning five model families: GPT-4 (OpenAI, 2023), GPT-2 (Radford et al., 2019), MPT-7B (MosaicML NLP Team, 2023), Mistral-7B-Instruct-v0.1 (Jiang et al., 2023), LLaMA-2-Chat (Touvron et al., 2023), and Cohere-Command. This covers two pure-base generators (GPT-2, MPT-7B), one SFT-only generator (Mistral-7B-Instruct-v0.1, fine-tuned on instruction data without RLHF (Jiang et al., 2023)), and three SFT+RLHF generators (GPT-4, LLaMA-2-Chat, Cohere-Command). We subsample 7,500 examples per generator (3,750 human, 3,750 AI) balanced across domains, yielding 45,000 samples in total.

3.2 Model and Activations

We use BERT-base-uncased (Devlin et al., 2019) with all weights frozen throughout. For each input text we extract the CLS-token hidden state at every transformer layer $l \in \{1, \ldots, 12\}$, giving a per-layer vector $\mathbf{h}^{(l)} \in \mathbb{R}^{768}$. The full hidden-state representation is the concatenation

$$\mathbf{z} = \left[\mathbf{h}^{(1)}; \ldots; \mathbf{h}^{(12)}\right] \in \mathbb{R}^{9,216}.$$

We refer to individual dimensions of $\mathbf{z}$ as *neurons*; neuron index i resides in layer $\lfloor i/768 \rfloor + 1$ and position $i \bmod 768$ within that layer. Using the frozen pretrained model (rather than a fine-tuned detector) ensures that any informative neurons we identify are present in general-purpose BERT representations, not artefacts of task-specific fine-tuning.

3.3 Protocol: L1 Selection, L2 Evaluation

Following Gurnee et al. (2023), for each (seed, fold) cell we fit two independent probes on the training split. (1) *L1 selection probe*: L1-regularised logistic regression on all 9,216 neurons, fitted with scikit-learn's `liblinear` solver at inverse regularisation strength $C = 0.005$ (smaller $C \Rightarrow$ stronger L1 penalty; see Appendix A for the exact loss). Neurons with non-zero coefficients form the *candidate set* $\mathcal{S}$; neurons appearing in $\mathcal{S}$ in at least 50% of the 15 cells are *stable neurons*. (2) *L2 evaluation probe*: L2-regularised logistic regression (lbfgs) on the full $\mathbf{z}$. This probe is held frozen for all downstream experiments to ensure that intervention effects are measured against a fixed decision function.

We use 5-fold cross-validation stratified by (label $\times$ domain) with seeds $\{42, 123, 456\}$, yielding 15 cells per (generator, experiment). All reported metrics are 15-cell means $\pm$ standard deviation unless otherwise noted.

Hyperparameter selection. $C = 0.005$ and $N = 7{,}500$ were chosen via a multi-generator stability sweep run prior to any results analysis, maximising the minimum-over-generators pairwise Jaccard subject to ≥ 50 neurons per draw (Gurnee et al., 2023). The chosen operating point achieves a minimum Jaccard of 0.676 (GPT-2 bottleneck; other five generators 0.707–0.751); full grid in Appendix B.

4 Sparse Detection Circuits in Frozen BERT

Table 1 reports sparse-probe results across all six generators. Between 45 and 62 neurons (0.49–0.67% of the 9,216-dimension CLS representation) are stable across all 15 cells, yet the L2 evaluation probe achieves 90.5–98.8% validation accuracy (AUC-ROC 0.968–0.999). A probe restricted to the stable set alone achieves 86.5–97.2% (see Appendix E). This confirms that the stable neurons are themselves sufficient for most of the detection accuracy. Selection is highly reproducible: pairwise fold Jaccard ranges 0.67–0.76 across 105 fold-pairs per generator, compared to a random-null expectation of $\sim$0.003–0.005 for sets of this size drawn from 9,216 neurons, placing within-generator stability at 130–200$\times$ above chance. The selection pool is narrow: between 91 neurons ever selected (LLaMA) and 146 (GPT-2). The probe thus consistently returns to a small region of the feature space. Layer-by-layer neuron distributions and the base/instruction-tuned asymmetry are examined in Section 6.

Generator	Val acc	AUC	n_{sel}	Jaccard	n_{stable} / n_{obs}
GPT-4	0.988 ± 0.003	0.999 ± 0.000	74.1 ± 2.9	0.763 ± 0.037	60 / 106
GPT-2	0.959 ± 0.003	0.992 ± 0.001	84.7 ± 4.1	0.670 ± 0.042	60 / 146
MPT	0.972 ± 0.003	0.994 ± 0.001	61.3 ± 3.1	0.740 ± 0.036	52 / 94
Mistral	0.971 ± 0.004	0.995 ± 0.001	57.8 ± 2.9	0.724 ± 0.046	45 / 99
LLaMA	0.986 ± 0.003	0.998 ± 0.001	63.6 ± 1.7	0.756 ± 0.040	48 / 91
Cohere	0.905 ± 0.008	0.968 ± 0.003	72.3 ± 3.1	0.749 ± 0.039	62 / 113

Table 1: Sparse-probe results across six generators: 15-cell means $\pm$ std (5 folds $\times$ 3 seeds). Val acc: L2 evaluation probe accuracy (all 9,216 neurons). AUC: area under the ROC curve. n_{sel}: mean candidate set size per cell. Jaccard: mean pairwise fold Jaccard over all 105 fold-pairs (C(15,2)), computed from per-fold selection sets. n_{stable}: neurons in $\geq$50% of cells; n_{obs}: total distinct neurons selected at least once.

Generator	Selected (%)	Random-k (%)	Ratio
GPT-4	$1.23_{\pm 0.33}$	$\sim$0.08	$\sim$16$\times$
GPT-2	$3.13_{\pm 0.54}$	$\sim$0.32	$\sim$10$\times$
MPT	$1.52_{\pm 0.22}$	$\sim$0.14	$\sim$11$\times$
Mistral	$1.73_{\pm 0.27}$	$\sim$0.12	$\sim$14$\times$
LLaMA	$1.07_{\pm 0.37}$	$\sim$0.11	$\sim$10$\times$
Cohere	$8.15_{\pm 1.27}$	$\sim$0.79	$\sim$10$\times$

Table 2: Forward-direction (AI $\rightarrow$ human) selected and random-k flip rates (%) at k = full (15-cell means $\pm$ std). Ratio = selected / random. All six generators fall in the 10–16$\times$ range. Full k-sweep in Appendix J. Reverse-direction (human $\rightarrow$ AI) results are reported in the text.

Per-generator CAV diagnostics (Appendix F) corroborate the cross-generator differences in §6: instruction-tuned generators show tighter signal geometry, while Cohere is the outlier throughout.

5 Causal Role of the Selected Neurons

Having identified a sparse set of neurons, we ask two complementary causal questions: are these neurons *sufficient* to produce AI-text predictions, and are they *necessary*? Section 5.1 addresses both via bidirectional activation patching—transplanting AI activations into human targets (forward; sufficiency) and human activations into AI targets (reverse; partial necessity); Section 5.2 complements this with mean ablation. A restricted-probe localisation analysis reinforcing the base/instruction-tuned split is provided in Appendix E.

5.1 Causal Role via Bidirectional Activation Patching

5.1.1 Patching Protocol

We run patching **bidirectionally** to assess both causal directions: (i) **forward** (AI donor $\rightarrow$ human target): transplanting the selected neurons' activations from AI samples into human targets, testing whether the selected set is *sufficient* to flip the probe to an AI prediction; and (ii) **reverse** (human donor $\rightarrow$ AI target): transplanting human activations into AI targets, testing whether the selected set is partially *necessary* for maintaining AI predictions.

For each (seed, fold) cell and direction we identify *valid target* samples: test-fold samples of the target class that share a domain with at least one test-fold donor sample. For each of $n_{\text{shuffles}} = 20$ random within-domain donor permutations we construct a patched activation by replacing the selected-neuron activations of the target with those of the donor ($\tilde{\mathbf{z}}_{\mathcal{S}} \leftarrow \mathbf{z}_{\mathcal{S}}^{\text{donor}}$), leaving non-selected neurons unchanged. A *flip* occurs when the frozen evaluation probe's predicted class changes after patching (human $\rightarrow$ AI in the forward direction; AI $\rightarrow$ human in the reverse). The flip rate is the fraction of valid target samples that flip, averaged over the 20 donor permutations and then over the 15 cells. Donor sampling is restricted to the same domain as the target, following the on-manifold recommendation of Heimersheim and Nanda (2024) to avoid injecting out-of-distribution activations.

The *random-k baseline* draws 20 random subsets of size k and runs the identical procedure, providing a null that controls for the number of patched dimensions. We sweep $k \in \{1, 5, 10, 20, 50, \text{ full}\}$, where k = full refers to the per-fold L1-selected set (53–92 neurons depending on generator and fold). The cross-fold *stable set* (neurons appearing in $\geq$50% of cells) is used for cross-generator analysis and LOGO evaluation, but not as the intervention target.

5.1.2 Flip-Rate Results (Forward and Reverse)

Forward (AI $\rightarrow$ human). Figure 2 and the full k-sweep (Table 13, Appendix J) show monotonically increasing flip rates across all generators. At k = full, patching the selected neurons causes 1.07–8.15% of human samples to be reclassified as AI-generated. Against a random baseline of 0.08–0.79% at the same k, this yields a ratio of 10–16$\times$ across all six generators. MPT falls at the lower end of the range (1.52%, $\sim$11$\times$ above random), consistent with its relatively diffuse cross-layer stable set (§6.1).

Generator-level differences track baseline accuracy: Cohere flips most readily (8.15%, lowest baseline at 90.5%), LLaMA least (1.07%, baseline 98.6%), consistent with proximity to the deci-

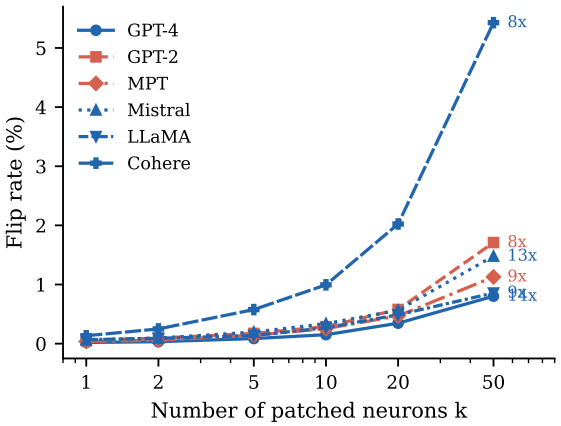

Figure 2: Forward flip rate (%) vs. number of patched neurons k on a log-scale x-axis, averaged over 15 cells (5 folds $\times$ 3 seeds). Instruction-tuned generators are shown in blue, base generators in red. Numbers to the right of each line are selected/random ratios at $k = 50$ (8–14$\times$); headline ratios at $k =$ full (the per-fold L1-selected set, 53–92 neurons) are 10–16$\times$ across generators (Table 13).

Generator	Base acc	Sel. drop	Rand. drop
GPT-4	0.988	$+0.03 \pm 0.12$ pp	$+0.01$ pp
GPT-2	0.959	$+0.11 \pm 0.31$ pp	$+0.00$ pp
MPT	0.972	$+0.05 \pm 0.11$ pp	$+0.00$ pp
Mistral	0.971	$+0.10 \pm 0.21$ pp	$+0.00$ pp
LLaMA	0.986	$+0.11 \pm 0.14$ pp	$+0.02$ pp
Cohere	0.905	$\mathbf{+1.06 \pm 0.52}$ pp	$+0.06$ pp

Table 3: Mean-ablation accuracy drop at $k =$ full, 15-cell means $\pm$ std (positive = drop). Sel. drop: ablating the stable set; Rand. drop: mean over 20 random k-subsets. Cohere is the only generator substantially above the random baseline.

sion boundary. Flip rate grows sub-linearly with k across all generators (Figure 2), inconsistent with a concentration of effect in a few critical units.

Reverse (human $\rightarrow$ AI). Transplanting human-donor activations into AI-target samples tests whether the selected neurons are partially necessary for maintaining AI-class predictions. At $k =$ full, 0.65–5.74% of AI samples were reclassified as human (per-generator breakdown in Table 13, Appendix J), consistently 7–12$\times$ above the random-neuron baseline. Reverse flip rates are lower than forward (0.65–5.74% vs. 1.07–8.15%), consistent with the partial-redundancy reading: the remaining 9,000+ unpatched neurons carry enough redundant AI signal to partially preserve the prediction even when the selected set is overwritten. The same-domain donor constraint eliminates domain shift by construction; mean-ablation caveats are discussed in §5.2.

5.2 Necessity via Mean Ablation

To complement the sufficiency test we ask the converse question: is the selected set *necessary*? We answer this via mean ablation, replacing the selected neurons in the test-fold activations with their training-fold mean (computed on the training split to avoid leakage) and re-evaluating the frozen L2 probe. Table 3 reports accuracy drop at $k =$ full alongside the random-k baseline.

For five of six generators the result is negative. At $k =$ full, ablating the selected set causes accuracy to drop by only 0.03–0.11 pp, within noise of the 0.00–0.06 pp random-k baseline; the drops for GPT-4, GPT-2, MPT, and Mistral are within one standard deviation of zero (Table 3), so no reliable necessity is detectable for these generators. The selected neurons are not individually necessary: the L2 evaluation probe, which uses all 9,216 neurons, retains thousands of redundant correlates that compensate for the removed subset.

Cohere is the exception. Its full-set ablation drop is 1.06 ± 0.52 pp, roughly 18$\times$ the random baseline of 0.06 pp, and the drop grows monotonically from $k = 5$ upward, unlike the other five generators where the curve is flat. Cohere's other signals point the same way: it has the lowest baseline accuracy, smallest mean-difference norm, and largest probe weight norm, all indicating a detection representation that is *partially* localised rather than fully distributed.

We treat the necessity result as a conservative lower bound: ablating $\leq$1% of a 9,216-dimensional probe is intrinsically weak, and mean ablation moves activations off-manifold (Heimersheim and Nanda, 2024); the reverse patching experiment (§5.1.2) provides an on-manifold necessity test confirming partial necessity. We therefore characterise the selected set not as *the* detection circuit but as a causally sufficient, partially necessary projection of a redundantly distributed signal — one of plausibly many sufficient subspaces, distinguished by being stably recoverable across folds, seeds, and (via LOGO) generator families.

6 Cross-Generator Circuit Geometry

6.1 Layer Distribution of Stable Neurons

Figure 3 shows the layer-by-layer distribution of stable neurons across all six generators. A clear asymmetry emerges between the four instruction-tuned generators and the two pure-base generators. For GPT-4, Mistral, LLaMA, and Cohere, the single most populated layer is layer 12, which accounts for 30–36% of each generator's stable set (16–19 neurons). Both pure-base generators are exceptions: GPT-2 peaks at layer 11 with 11 stable neurons (18.3% of its stable set) and has only 13.3% in layer 12; MPT peaks earlier with just 2 stable neurons in layer 12 (3.8%) and distributes its signal more broadly across earlier layers (layers 1, 8, and 9 dominate).

Across both pure-base generators, the shared pattern is markedly low layer-12 concentration (3.8–13.3%) versus any instruction-tuned generator (30–36%). This consistent L12-avoidance (not a rigid L11-peaking rule: GPT-2 peaks at L11 while MPT peaks at L1/L8) is consistent with BERT's final layer behaving as a classification-register-like site that is preferentially engaged by instruction-tuned outputs and bypassed by base-model outputs (Tenney et al., 2019; Rogers et al., 2020; Geva et al., 2021). We stress that this reading rests on $N = 2$ base generators (GPT-2, MPT) and is descriptive of the selected-neuron geometry rather than a mechanistic claim about RLHF/SFT-induced circuitry; we treat it as a working hypothesis (see §9).

6.2 Stable-Set Overlap Across Generators

Pairwise Jaccard similarity between stable sets (Table 12 in Appendix I) ranges from 0.00 to 0.24, confirming generator-specific circuits. However, the appropriate baseline is not zero but the expected Jaccard for two random sets of the same sizes drawn from 9,216 neurons, which is approximately 0.003 per pair. Relative to this null, the mean observed Jaccard of 0.073 is $\sim$24$\times$ above chance, which indicates a partial shared substrate despite the generator-specific structure.

The data reveal a clear bipartite structure (Figure 4 in Appendix I). Among instruction-tuned generators, pairwise Jaccard ranges 0.08–0.24 (27–86$\times$ chance), with the strongest pairs being Cohere–Mistral (86$\times$) and LLaMA–Mistral (64$\times$). Among the two pure-base generators, MPT and GPT-2 share 12 neurons (J = 0.120, 40$\times$ chance), more than several instruction–instruction pairs. This

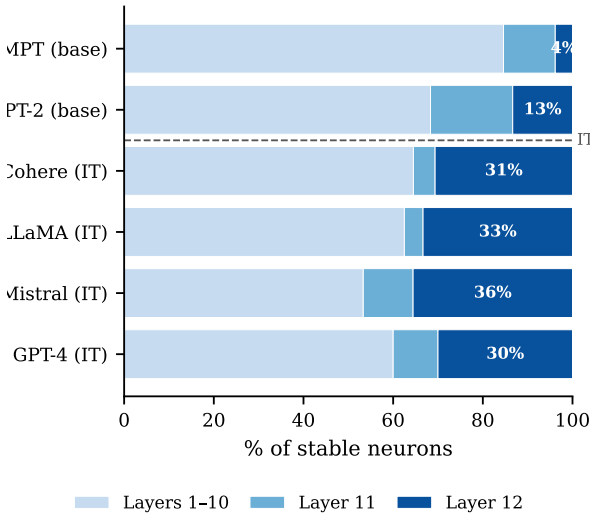

Figure 3: Proportion of stable neurons in each layer group per generator. Layer 12 percentage annotated in white. Instruction-tuned generators (IT) concentrate 30–36% of stable neurons in layer 12; base generators retain $\leq$14%. Dashed line separates the two groups.

suggests the base models form a coherent subcluster. In contrast, every base-vs-instruction-tuned pair has near-zero overlap (J $\leq$ 0.05), with three pairs at exactly zero (full breakdown in Table 12). This base-vs-instruction-tuned partition explains the bulk of the cross-generator circuit geometry.

6.3 A Cross-Generator Core Concentrated in Layer 12

Neurons appearing in the stable sets of at least 3 of the 6 generators (minimal majority threshold) form a *core* set of 17 neurons. Of these, 10 (59%) reside in layer 12, far exceeding any individual generator's L12 share (3.8–36%); the remaining 7 are scattered across layers 1, 2, 5, 7, 8, and 11. This concentration is the clearest descriptive evidence that BERT's layer-12 representations form the primary cross-generator site for AI-detection signal: a majority of neurons shared across half or more of the studied generators converge there, even though no individual generator devotes a majority of its own stable set to L12. Each generator's remaining stable neurons form a generator-specific peripheral set, consistent with the bipartite Jaccard geometry of §6.2 and with the residual Sel/Full gap observed in the LOGO evaluation (§7).

7 Leave-One-Family-Out Generalisation

The experiments above characterise stable neurons for each generator *independently*. A stronger test

of the mechanistic claim is whether a set of neurons identified from *other* generators transfers to a held-out generator family never seen during neuron selection. We address this with a leave-one-family-out (LOGO) evaluation.

Protocol. We partition all 11 RAID generators into five families: `gpt`, `cohere`, `llama`, `mistral`, and `mpt`. For each held-out family we run the sparse-probe pipeline on the remaining generators and evaluate two probes on the held-out set: (i) **Full L2** on all 9,216 neurons, and (ii) **Sel L2** on only the stable neurons identified from training families. To handle the larger combined training pool, LOGO uses a tighter regularisation ($C = 0.001$) and caps each generator's contribution at 5,000 samples (compared to $C = 0.005$, $N = 7{,}500$ in the per-generator headline runs). Family sizes are unequal (1–4 generators per family, following RAID's taxonomy), so training pool size and test-set size vary across folds. The Sel/Full ratio in Table 4 normalises for this by expressing each family's held-out accuracy as a fraction of its own full-feature ceiling.

Results. Table 4 reports the outcome. Full L2 generalises strongly in all five conditions (88.6–99.7% accuracy), showing that frozen BERT representations are themselves informative across the RAID generator space. The key finding is in the Sel L2 column: 91–125 stable neurons (at most 1.4% of 9,216 neurons) achieve 76.3–93.7% accuracy on generators from entirely held-out families. Expressed as a fraction of the full-feature ceiling for each family, the selected set retains 86–94% of the full-feature signal (Table 4, Sel/Full column).

The worst-case family is `cohere` (Sel L2 = 76.3%, 86% of ceiling), as expected from its low pairwise Jaccard with other generators; the best is `llama` (93.7%, 94%), which benefits from high overlap within the instruction-tuned group (§6.2). `mpt` and `mistral` fall in the middle tier (85.6–85.7%).

The 6–14 pp residual gap between Sel L2 and Full L2 has a natural mechanistic reading: a cross-family core of neurons carries most of the detection signal that transfers, while the remainder of the full-feature ceiling is supplied by *family-specific* neurons that the LOGO procedure cannot observe by construction. This is consistent with the bipartite Jaccard structure (§6.2), where each generator's stable set is dominated by a generator-specific periphery layered on a small shared substrate.

Family	n_{stable}	Full L2	Sel L2	Sel/Full	Full AUC	Sel AUC
cohere	119	88.6%	76.3%	86%	0.946	0.863
gpt	91	98.2%	88.8%	90%	0.993	0.953
llama	125	99.7%	93.7%	94%	0.999	0.986
mistral	106	96.6%	85.6%	89%	0.982	0.925
mpt	110	99.1%	85.7%	87%	0.995	0.926

Table 4: Leave-one-family-out (LOGO) generalisation. n_{stable}: stable neurons identified from the four training families. Full L2: L2-probe accuracy using all 9,216 BERT neurons. Sel L2: L2-probe accuracy restricted to n_{stable} neurons. Sel/Full: ratio expressing how much of the full-feature ceiling the selected neurons retain. Full AUC: area under the ROC curve (AUC) for the full-feature L2 probe. Sel AUC: AUC for the Sel L2 probe, ranging from 0.863 (cohere) to 0.986 (llama). All metrics are means over 3 seeds $\times$ 5 folds = 15 cells. Test set sizes: 5,000 (llama, 1 generator) to 20,000 (gpt, 4 generators).

8 Conclusion

An L1-regularised probe stably selects 45–62 neurons ($<$1% of 9,216 CLS dimensions) across six RAID generators. Bidirectional activation patching shows these neurons are both causally sufficient (forward, 10–16$\times$ above random) and partially necessary (reverse, 7–12$\times$ above random), while same-set mean ablation costs $\leq$1.1 pp accuracy on five of six generators. The joint pattern — large flip rates alongside negligible ablation drops — is the signature of a signal that is *easy to inject, hard to erase*: redundantly distributed across the full representation but concentrated enough to be causally sufficient, consistent with the broader redundancy phenomenon documented in pretrained transformers (Dalvi et al., 2020).

Cross-generator analysis reveals a bipartite structure: instruction-tuned generators concentrate stable neurons in BERT's final layer while both pure-base generators do not, suggesting post-training alignment leaves a footprint in layer 12. Cohere sits within rather than against this reading — its diffuse geometry (lowest CAV cosine alignment, largest probe weights, only generator with a substantial ablation cost) reflects an instruction-tuned generator encoding the same layer-12 signal at lower per-neuron magnitude, which the probe compensates for with larger weights. Leave-one-family-out evaluation confirms that neurons identified from training families retain 86–94% of the full-feature ceiling on entirely held-out generator families, providing evidence that the discovered circuits are not artefacts of the specific generators studied while

the residual gap leaves room for family-specific neurons beyond the cross-family core.

9 Limitations

Single encoder architecture. All experiments use frozen BERT-base-uncased. Whether the stable-neuron structure and patching effects extend to larger encoders (RoBERTa-large (Liu et al., 2019), DeBERTa-v3 (He et al., 2023)) or encoder-decoder architectures is an open question. The frozen setting is a deliberate design choice: it isolates what the pretrained representation already encodes, but it means results cannot be directly compared to fine-tuned detectors without controlling for this variable.

English-only evaluation. We evaluate exclusively on the English-language RAID benchmark. Whether the layer-12 footprint of post-training alignment, the bipartite base/instruction-tuned circuit structure, and the redundancy patterns we observe generalise to languages with richer morphology, different syntactic structure (e.g., agglutinative or pro-drop languages), or non-Latin scripts requires separate evaluation. Multilingual encoder variants (e.g., mBERT, XLM-R) would be a natural starting point.

Mean ablation is off-manifold. Replacing selected neurons with their training-mean pushes activations off the natural data manifold, which can introduce artefacts unrelated to the detection signal (Heimersheim and Nanda, 2024). We treat the necessity results as conservative lower bounds; a zero-ablation comparison and full resample-ablation analysis remain future work.

Layer-asymmetry claim across two base generators. The observation that instruction-tuned generators concentrate 30–36% of stable neurons in layer 12 while pure-base generators have $\leq$14% is now supported by two independent base models (GPT-2 and MPT-7B). However, the two base generators differ in their peak layer (GPT-2: L11; MPT-7B: L1/L8), so the claim is best stated as L12-avoidance rather than a strict L11-peaking rule. We frame this as a consistent L12 footprint of post-training alignment rather than a mechanistically precise claim about which specific earlier layer is activated.

Adversarial robustness not evaluated. RAID includes eleven adversarial rephrasing attacks (paraphrase, homoglyph, synonym substitution, etc.) that substantially reduce detector accuracy (Dugan et al., 2024). Whether the stable neurons identified here remain causal under these attacks is left to future work; the present analysis covers only the non-adversarial RAID subset.

Scope of causal claim. Activation patching provides evidence that the selected neurons are sufficient to shift probe predictions, but does not prove causality at the circuit level (Meng et al., 2022). The patching intervention is at the level of individual neuron activations, not a complete mechanistic decomposition of the computation. We make no claims about what linguistic properties these neurons encode or why they are informative.

Ethical considerations

AI-text detection is a dual-use capability: detectors can support platform integrity and journalistic verification, but can equally be used to penalise legitimate AI-assisted writing or to guide adversarial evasion. This work is mechanistic-interpretability research: we study which neurons drive an existing frozen encoder's representations, not deploy a new detection system. All data come from the publicly released RAID benchmark (Dugan et al., 2024); no human subjects were involved and no personally identifiable information was collected or processed. The stable neuron sets we identify could in principle inform evasion strategies (by targeting the identified circuits), but the same information could equally guide more robust detector design. We release all experiment code to support reproducibility.

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A Pipeline Details

Figure 1 in Section 1 provides a visual overview of the pipeline described below.

Data sampling. For each generator we draw $N = 7{,}500$ balanced samples from the RAID benchmark: equal numbers of AI-generated and human-written texts, stratified across the six RAID domains (abstracts, books, news, Reddit, reviews, Wikipedia) using stratified sampling without replacement. Texts shorter than 50 whitespace-delimited tokens are excluded. Human texts for each generator are the original source passages from which the AI texts were generated; this pairs each AI sample with a human sample from the same domain and source.

Activation extraction. We encode each text with BERT-base-uncased using HuggingFace Transformers (Wolf et al., 2020). The input is truncated to 512 WordPiece tokens. We extract the CLS-token hidden state from *all* 12 transformer layers and concatenate them into a single vector $\mathbf{z} \in \mathbb{R}^{9{,}216}$ (12×768). Gradients are disabled throughout; BERT weights are never updated.

Normalisation. Before any probe training, each of the 9,216 neurons is z-scored with a `StandardScaler` fit on the training fold and applied to the test fold. The same scaler is reused for all downstream experiments (patching, ablation) within a given cell so that scale is held constant across interventions.

Cross-validation protocol. We use 5-fold stratified cross-validation repeated with 3 random seeds (42, 123, 456), giving 15 independent evaluation cells per generator. The stratification is over the binary AI/human label; domain balance is not enforced at the fold level but is approximately preserved because the initial sample is domain-stratified.

L1 selection. The *selector* is a ℓ_1-regularised logistic regression (`liblinear` solver, $C = 0.005$, `max_iter=` 1,000, `class_weight=balanced`) fit on the training fold. Concretely, `liblinear` minimises

$$\tfrac{1}{2C}\|\mathbf{w}\|_1 + \sum_{i=1}^{n} \log\big(1 + \exp(-y_i\, \mathbf{x}_i^\top \mathbf{w})\big)$$

over training-fold samples $(\mathbf{x}_i, y_i)$ with $y_i \in \{-1, +1\}$; smaller C inflates the ℓ_1 coefficient and therefore enforces a sparser solution. The selected set $\mathcal{S}$ for a given cell is the set of neurons with non-zero weight. All 9,216 neurons are provided as input; the regulariser drives most to zero.

Stable neuron aggregation. A neuron is *stable* if it appears in $\mathcal{S}$ in at least 50% of the 15 cells (≥ 8 out of 15). This threshold was chosen to require cross-seed and cross-fold consistency simultaneously. Neurons appearing in 0–7 cells are discarded; those in 8–15 cells form the stable set $\mathcal{S}^*$.

L2 evaluation probe. The *evaluator* is a separate ℓ_2-regularised logistic regression (`lbfgs` solver, $C = 1.0$, `max_iter=` 1,000) trained and tested on each fold using the same scaler. For the main sparse-probe accuracy (Table 1), the L2 probe is trained on all 9,216 neurons. For patching and ablation experiments, it is trained on a restricted neuron set (either stable neurons only, or the full set with selected neurons patched/ablated). The L2 probe is retrained for each fold using only the fold's training split; the scaler is not refit.

Patching protocol. Patching uses same-domain donor matching: for each human target sample, a donor AI sample from the same domain is drawn without replacement. The selected k neuron activations of the target are replaced with the corresponding values from the donor. The k neurons are always the first k members of $\mathcal{S}^*$ sorted by their frequency of appearance across cells (most stable first). For the full-k sweep, all $|\mathcal{S}^*|$ neurons are patched.

Software and licensing. We use the HuggingFace Transformers library (Wolf et al., 2020) for BERT inference and `scikit-learn` for all linear probes. RAID is released under the MIT license; BERT-base-uncased is released under the Apache 2.0 license. We use both within their permitted scope (non-commercial research analysis); no data or model weights are redistributed.

Computational budget. BERT-base-uncased has 110M parameters and is used in inference-only mode (no fine-tuning) throughout. CLS activations for all 45,000 samples are encoded once and cached; all downstream experiments (L1/L2 probes, stability sweep, mean ablation, bidirectional patching k-sweep, LOGO evaluation) operate on the cached representations. The complete pipeline runs in approximately 7–8 wall-clock hours on a consumer workstation: a single CUDA-capable GPU is used only for the one-off BERT activation-encoding pass ($\sim$30 minutes for all 45,000 samples), while the multi-generator stability grid ($\sim$1 hour), the six

per-generator main runs including the k-sweep with $n_{\text{shuffles}} = 20$ donor permutations ($\sim$5 hours total), and the LOGO evaluation across five family folds ($\sim$15 minutes) are all CPU-bound, since L1/L2 logistic regression, mean ablation, and activation patching with cached representations do not benefit from GPU acceleration. The pipeline is therefore I/O- rather than compute-bound after the initial encoding pass.

B Stability Sweep

Section 3.3 justifies the choice of $C = 0.005$ and $N = 7{,}500$ by reference to this appendix. The stability sweep evaluates mean pairwise Jaccard similarity between selection sets produced by 15 independent draws of the L1 selector (varying random seed and fold assignment) for all six RAID generators across a grid of regularisation strengths $C \in \{0.001, 0.002, 0.005, 0.01, 0.02, 0.05\}$ and sample sizes $N \in \{500, 1000, 2000, 3500, 5000, 7500\}$ ($K = 15$ draws per cell). Higher Jaccard means the selection is more reproducible. Values in parentheses are the mean number of neurons selected per draw $\bar{n}$.

Three observations drive the choice of $C = 0.005$, $N = 7{,}500$. **(1) Sparse-end cut-off:** $C \leq 0.002$ yields high or trivial Jaccard at small N but selects fewer than 22 neurons at $N = 7{,}500$ across all generators—too sparse for circuit analysis. Among operating points selecting ≥ 50 neurons per draw, $C = 0.005$ achieves the highest minimum-over-6-generators Jaccard. **(2) Regularisation:** at any fixed $N \geq 2{,}000$ in the $C \geq 0.005$ region, $C = 0.005$ dominates all weaker regularisation values (larger C) on every generator. **(3) Sample size:** for $C = 0.005$ Jaccard increases monotonically in N, with the largest gains between $N = 1{,}000$ and $N = 3{,}500$. The $N = 5{,}000 \rightarrow 7{,}500$ step adds 0.016–0.103 per generator; $N = 10{,}000$ cannot be evaluated because the pool is exhausted ($K = 1$). The chosen operating point achieves a minimum-over-6-generators Jaccard of 0.676, with GPT-2 as the bottleneck (0.676) and the remaining five generators ranging 0.707–0.751. **(4) Degenerate region:** at $N \leq 500$ and $C = 0.005$, every generator selects zero neurons (Jaccard $= 1.0$ trivially); $N = 7{,}500$ reliably produces non-empty sets across all six generators.

C Per-Cell Metrics

Table 6 reports L2-probe accuracy, AUC-ROC, F1, and number of L1-selected neurons for every evaluation cell (seed $\times$ fold).

D AUC vs L1 Selection

We compare L1-regularised selection (the method used throughout this paper) to a univariate AUC-based ranking: for each neuron, compute the one-vs-rest AUC-ROC against the binary label on the training fold, and retain the top-k neurons where k is chosen to match the L1 count for a fair comparison. We evaluate the informativeness of each selected set by *complement ablation*: the L2 probe is re-evaluated after zero-ing out all neurons *not* in the selected set (i.e., only the selected neurons are retained). A smaller drop from the full-probe baseline indicates the selected set is more self-contained.

The Jaccard overlap between L1 and AUC-selected sets is low (0.10–0.19) despite both being applied to the same data. The two criteria thus select largely non-overlapping neurons. For GPT-4, Mistral, and LLaMA, L1 achieves the same or better task performance with $4\times$–$6\times$ fewer neurons: ablating L1's 58–74 neurons costs only 0.03–0.11pp, while ablating AUC's 238–491 neurons costs 0.52–1.02pp. This parsimony makes L1 preferable for mechanistic analysis: a smaller identified set is easier to interpret and less likely to include redundant or near-duplicated neurons.

For GPT-2, the two methods select nearly the same number of neurons (85 vs. 86) with similar ablation cost; neither dominates. For Cohere, AUC selects *fewer* neurons (38 vs. 72) and its ablation cost is lower (+0.28pp vs. +1.06pp): Cohere's detection signal is concentrated in a small number of individually highly-discriminative neurons that univariate ranking recovers more directly than joint L1 regularisation. We discuss this reversal in the main text (Section 6.1): it fits Cohere's distinct geometry, since the model with the lowest baseline probe accuracy is also the one where AUC beats L1 on parsimony.

E Signal Localisation: Restricted Probe

The necessity result (§5.2) uses the full-feature L2 probe, which retains thousands of redundant correlates. A complementary question is whether the stable neurons alone are sufficient for a *freshly trained*

Table 5: Mean Jaccard stability (and mean $\bar{n}$ selected) across 15 random draws of the L1 selector. Rows are C values; columns are N values. Larger Jaccard is better. **Bold**: chosen operating point ($C = 0.005$, $N = 7{,}500$). Dagger (†): degenerate cell ($\bar{n} < 1$, Jaccard trivially 1.0 or unstable). $N = 10{,}000$ omitted (only $K = 1$ draw possible; pool exhausted).

C	$N = 500$	$N = 1000$	$N = 2000$	$N = 3500$	$N = 5000$	$N = 7500$
GPT-4						
0.001	1.000† (0)	1.000† (0)	1.000† (0)	0.867 (0)	0.689 (8)	0.808 (18)
0.002	1.000† (0)	1.000† (0)	0.680 (3)	0.664 (16)	0.692 (28)	0.799 (44)
0.005	1.000† (0)	0.433 (9)	0.510 (29)	0.617 (51)	0.684 (65)	**0.751 (80)**
0.01	0.314 (9)	0.356 (29)	0.488 (54)	0.553 (78)	0.598 (97)	0.681 (123)
0.02	0.254 (29)	0.355 (53)	0.425 (82)	0.460 (117)	0.509 (143)	0.625 (179)
0.05	0.205 (57)	0.282 (88)	0.310 (134)	0.353 (184)	0.414 (232)	0.535 (283)
GPT-2						
0.001	1.000† (0)	1.000† (0)	1.000† (0)	1.000† (0)	0.579 (2)	0.932 (14)
0.002	1.000† (0)	1.000† (0)	1.000† (0)	0.572 (12)	0.744 (19)	0.833 (37)
0.005	1.000† (0)	0.288 (3)	0.464 (20)	0.530 (49)	0.581 (70)	**0.676 (103)**
0.01	0.197 (3)	0.308 (22)	0.385 (55)	0.443 (97)	0.512 (131)	0.660 (183)
0.02	0.202 (21)	0.260 (53)	0.304 (106)	0.405 (169)	0.463 (216)	0.574 (289)
0.05	0.133 (64)	0.191 (118)	0.241 (204)	0.306 (307)	0.357 (388)	0.489 (505)
MPT						
0.001	1.000† (0)	1.000† (0)	1.000† (0)	0.467 (1)	0.861 (7)	0.840 (14)
0.002	1.000† (0)	1.000† (0)	0.546 (3)	0.704 (13)	0.774 (21)	0.832 (33)
0.005	1.000† (0)	0.496 (8)	0.619 (21)	0.624 (38)	0.638 (53)	**0.707 (78)**
0.01	0.341 (7)	0.437 (22)	0.463 (42)	0.455 (74)	0.529 (99)	0.666 (130)
0.02	0.281 (21)	0.300 (43)	0.339 (84)	0.409 (125)	0.448 (157)	0.565 (210)
0.05	0.164 (52)	0.206 (90)	0.241 (151)	0.294 (227)	0.347 (296)	0.477 (377)
Mistral						
0.001	1.000† (0)	1.000† (0)	1.000† (0)	1.000† (0)	0.705 (7)	0.867 (19)
0.002	1.000† (0)	1.000† (0)	0.214 (1)	0.714 (17)	0.731 (27)	0.860 (33)
0.005	1.000† (0)	0.435 (8)	0.552 (26)	0.620 (38)	0.684 (52)	**0.718 (78)**
0.01	0.259 (7)	0.461 (25)	0.450 (45)	0.475 (71)	0.524 (98)	0.632 (145)
0.02	0.322 (24)	0.317 (44)	0.333 (82)	0.357 (133)	0.424 (178)	0.546 (239)
0.05	0.171 (53)	0.190 (96)	0.220 (165)	0.288 (248)	0.345 (316)	0.482 (415)
LLaMA						
0.001	1.000† (0)	1.000† (0)	1.000† (0)	1.000† (0)	0.703 (7)	0.816 (22)
0.002	1.000† (0)	1.000† (0)	0.783 (2)	0.637 (18)	0.782 (30)	0.840 (39)
0.005	1.000† (0)	0.464 (7)	0.559 (30)	0.681 (44)	0.670 (53)	**0.748 (78)**
0.01	0.291 (7)	0.444 (27)	0.519 (48)	0.541 (72)	0.618 (96)	0.696 (120)
0.02	0.280 (23)	0.380 (46)	0.406 (78)	0.452 (114)	0.530 (147)	0.616 (188)
0.05	0.208 (49)	0.259 (86)	0.291 (137)	0.345 (201)	0.417 (249)	0.534 (310)
Cohere						
0.001	1.000† (0)	1.000† (0)	1.000† (0)	1.000† (0)	0.692 (1)	0.814 (13)
0.002	1.000† (0)	1.000† (0)	1.000† (0)	0.568 (10)	0.708 (21)	0.762 (35)
0.005	1.000† (0)	0.137 (2)	0.469 (21)	0.553 (42)	0.612 (57)	**0.715 (88)**
0.01	0.129 (2)	0.309 (19)	0.370 (50)	0.414 (85)	0.476 (117)	0.579 (172)
0.02	0.191 (21)	0.211 (51)	0.263 (102)	0.310 (164)	0.364 (225)	0.497 (327)
0.05	0.101 (61)	0.127 (122)	0.165 (226)	0.217 (352)	0.285 (472)	0.443 (642)

Table 6: Per-cell L2-probe metrics for all six generators. Each block of 15 rows is one generator; columns are seed and fold index. Acc: accuracy; AUC: AUC-ROC; n_{sel}: neurons selected by L1 in this cell.

Gen.	Seed	Fold	Acc	AUC	n_{sel}
GPT-4	42	0	0.9887	0.9991	74
	42	1	0.9853	0.9992	73
	42	2	0.9887	0.9991	80
	42	3	0.9853	0.9989	74
	42	4	0.9927	0.9989	70
	123	0	0.9860	0.9987	70
	123	1	0.9880	0.9992	76
	123	2	0.9860	0.9989	73
	123	3	0.9907	0.9995	73
	123	4	0.9867	0.9991	75
	456	0	0.9907	0.9989	77
	456	1	0.9907	0.9994	75
	456	2	0.9907	0.9991	72
	456	3	0.9840	0.9981	71
	456	4	0.9847	0.9993	79
GPT-2	42	0	0.9587	0.9923	80
	42	1	0.9587	0.9907	81
	42	2	0.9593	0.9906	80
	42	3	0.9593	0.9918	82
	42	4	0.9613	0.9909	90
	123	0	0.9627	0.9912	90
	123	1	0.9593	0.9912	82
	123	2	0.9647	0.9931	84
	123	3	0.9513	0.9908	86
	123	4	0.9613	0.9916	80
	456	0	0.9573	0.9929	90
	456	1	0.9580	0.9914	84
	456	2	0.9547	0.9911	82
	456	3	0.9620	0.9931	92
	456	4	0.9580	0.9902	87
MPT	42	0	0.9700	0.9939	59
	42	1	0.9720	0.9931	60
	42	2	0.9733	0.9954	60
	42	3	0.9700	0.9947	61
	42	4	0.9800	0.9960	61
	123	0	0.9687	0.9908	56
	123	1	0.9700	0.9961	68
	123	2	0.9780	0.9951	60
	123	3	0.9700	0.9934	65
	123	4	0.9700	0.9949	60
	456	0	0.9740	0.9938	65
	456	1	0.9700	0.9941	66
	456	2	0.9740	0.9939	59
	456	3	0.9747	0.9942	61
	456	4	0.9680	0.9931	59

Gen.	Seed	Fold	Acc	AUC	n_{sel}
Mistral	42	0	0.9727	0.9952	59
	42	1	0.9700	0.9950	53
	42	2	0.9607	0.9946	62
	42	3	0.9713	0.9954	56
	42	4	0.9780	0.9961	58
	123	0	0.9720	0.9945	60
	123	1	0.9667	0.9929	61
	123	2	0.9753	0.9966	55
	123	3	0.9700	0.9939	55
	123	4	0.9747	0.9954	62
	456	0	0.9733	0.9967	60
	456	1	0.9667	0.9935	54
	456	2	0.9747	0.9949	60
	456	3	0.9693	0.9961	57
	456	4	0.9660	0.9937	55
LLaMA	42	0	0.9920	0.9982	63
	42	1	0.9833	0.9973	64
	42	2	0.9900	0.9989	63
	42	3	0.9813	0.9990	64
	42	4	0.9827	0.9981	64
	123	0	0.9900	0.9995	65
	123	1	0.9847	0.9970	63
	123	2	0.9840	0.9988	62
	123	3	0.9880	0.9993	61
	123	4	0.9853	0.9980	64
	456	0	0.9840	0.9977	67
	456	1	0.9853	0.9985	67
	456	2	0.9853	0.9986	62
	456	3	0.9900	0.9991	61
	456	4	0.9867	0.9985	64
Cohere	42	0	0.9107	0.9716	71
	42	1	0.8967	0.9673	72
	42	2	0.8960	0.9612	72
	42	3	0.9213	0.9710	73
	42	4	0.9053	0.9669	72
	123	0	0.8973	0.9632	65
	123	1	0.9060	0.9665	76
	123	2	0.9133	0.9692	72
	123	3	0.9013	0.9704	76
	123	4	0.9027	0.9681	72
	456	0	0.9107	0.9704	69
	456	1	0.9013	0.9671	73
	456	2	0.8980	0.9629	79
	456	3	0.9173	0.9729	70
	456	4	0.9000	0.9653	73

Table 7: L1 vs. AUC selection: mean across 15 cells. n: neurons selected. Jaccard: overlap between the two selection sets. Drop: accuracy drop from baseline when complement is ablated (positive = degradation). Three regimes are visible: parsimony (GPT-4, Mistral, LLaMA: L1 selects far fewer neurons with equivalent or smaller drop), convergence (GPT-2, similar n and similar drop), and reversal (Cohere: AUC selects fewer with smaller drop).

Gen.	n_{AUC}	n_{L1}	Jaccard	AUC drop	L1 drop
GPT-4	491	74	0.127	+1.02pp	+0.03pp
GPT-2	86	85	0.096	+0.06pp	+0.11pp
Mistral	238	58	0.148	+0.52pp	+0.10pp
LLaMA	308	64	0.139	+0.55pp	+0.11pp
Cohere	38	72	0.185	+0.28pp	+1.06pp

probe, and how this compares to ablating the complement. We measure the *localisation gap*: the accuracy difference between a probe retrained on only the $|\mathcal{S}|$ stable neurons (*smaller model*) and the original L2 probe evaluated with non-stable neurons mean-ablated (*ablate complement*). A large positive gap means the stable set can support an independent probe on its own, but the full-feature probe leaned on non-stable neurons to define its decision boundary.

Generator	Smaller model	Gap vs ablate-complement
GPT-4	0.972 ± 0.005	$+0.41 \pm 0.42$ pp
GPT-2	0.913 ± 0.006	$+3.63 \pm 0.81$ pp
MPT	0.944 ± 0.005	$+4.68 \pm 1.14$ pp
Mistral	0.934 ± 0.005	$+0.93 \pm 0.42$ pp
LLaMA	0.960 ± 0.003	$+1.14 \pm 0.43$ pp
Cohere	0.865 ± 0.009	$+4.43 \pm 1.02$ pp

Table 8: Restricted probe results. *Smaller model*: L2 probe retrained on only the $|\mathcal{S}|$ stable neurons. *Gap*: smaller-model accuracy minus ablate-complement accuracy (positive = stable neurons self-sufficient; larger = more dependence on non-stable neurons in the full-feature probe). 15-cell means ± std.

The pattern splits cleanly along the base/instruction-tuned axis. Instruction-tuned generators (GPT-4, Mistral, LLaMA) show gaps of $\leq$1.2 pp: their stable neurons are nearly self-sufficient and the full-feature probe does not rely heavily on non-stable dimensions. The two pure-base generators and Cohere show gaps of 3.6–4.7 pp: their full-feature probes lean more on non-stable neurons, making complement ablation more costly. This points to a more distributed encoding structure for these three generators, in line with the layer-12 avoidance seen in base generators and the large ablation sensitivity of Cohere documented in the main body.

F Detection Signal Geometry (CAV Diagnostics)

To characterise the geometry of the detection signal within each stable set we compute a concept activation vector (CAV) (Kim et al., 2018) as the mean difference between AI and human activations restricted to the stable neurons, and compare it to the L1 probe weight vector. Table 9 reports cosine similarity between the CAV and the probe weight, logistic-regression accuracy on the CAV projection, and the norms of both vectors.

Cohere has the lowest CAV cosine similarity (0.057 ± 0.001), the lowest CAV-LR accuracy (0.910 ± 0.008), and by far the largest probe weight norm (108.3 ± 2.1), indicating a diffuse detection signal that the probe compensates for with large weights. MPT-7B is a notable exception: it has a low cosine (0.065 ± 0.000, close to Cohere's) but the *largest* mean-difference norm (9.52), meaning the AI − human class separation is geometrically strong per-axis yet misaligned with the probe direction. GPT-2 also shows a low cosine (0.073 ± 0.002) and elevated weight norm (69.9 ± 2.9), placing both base generators at the low-cosine end. Instruction-tuned generators cluster at higher cosine (0.110–0.138) with the smallest weight norms: GPT-4 has the most compact weight vector (45.8 ± 1.7), suggesting its detection signal is both geometrically aligned and concentrated. Each metric is mean ± std across three LR train/test split seeds; the mean-difference norm is deterministic (no split randomness). These per-generator geometric differences align with the intervention asymmetries documented in §5.

Generator	Cosine	LR acc	$\|\Delta\mu\|$	$\|w\|$
GPT-4	0.138 ± 0.004	0.988 ± 0.003	7.34	$\mathbf{45.8} \pm 1.7$
GPT-2	0.073 ± 0.002	0.963 ± 0.006	6.84	69.9 ± 2.9
MPT	0.065 ± 0.000	0.973 ± 0.001	**9.52**	53.8 ± 2.7
Mistral	0.110 ± 0.001	0.965 ± 0.003	6.38	59.2 ± 2.3
LLaMA	0.128 ± 0.001	0.987 ± 0.003	6.57	48.3 ± 0.9
Cohere	$\mathbf{0.057} \pm 0.001$	$\mathbf{0.910} \pm 0.008$	4.69	108.3 ± 2.1

Table 9: CAV diagnostics on per-generator stable sets. All values mean ± std across 3 random seeds. Cosine: similarity between mean-difference CAV and L1 probe weight. LR acc: logistic-regression accuracy on the CAV projection. $\|\Delta\mu\|$: AI − human mean-difference norm (deterministic; bold = largest). $\|w\|$: L1 weight norm (bold = smallest).

G Per-Domain Patching Breakdown

Table 10 reports selected flip rates (%) at $k =$ full broken down by RAID domain, averaged over all 15 cells (5 folds $\times$ 3 seeds). Values are 15-cell means; per-cell standard deviations ranged 0.6–2.6 pp across all generator–domain combinations.

H Stable-Neuron Layer Distributions (Full Table)

Table 11 reports the per-layer distribution of stable neurons for all six generators, summarised in §6.1.

I Pairwise Jaccard Similarity (Full Table)

Table 12 reports pairwise Jaccard similarity between stable sets for all 15 generator pairs, discussed in §6.2. Figure 4 visualises the same data as a heatmap with generators reordered to expose the base/instruction-tuned partition.

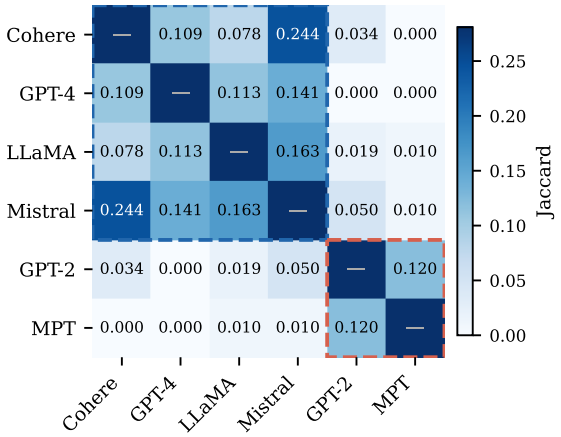

Figure 4: Pairwise Jaccard similarity heatmap (6×6). Generators reordered: instruction-tuned (IT) first, base last. Dashed blue box: IT×IT block. Dashed red box: base×base block. The bipartite structure (higher within-group similarity, near-zero cross-group overlap) is immediately visible.

J Full k-Sweep Flip Rates

Table 13 reports selected flip rates (%) across all k values for each generator, referenced in §5.1.2.

Domain	GPT-4	GPT-2	MPT	Mistral	LLaMA	Cohere
Abstracts	1.44	3.08	1.39	1.79	1.47	9.89
Books	1.08	1.93	1.47	0.92	1.21	10.06
News	1.61	4.92	2.74	2.11	1.01	9.67
Reddit	1.05	3.22	1.36	1.34	0.76	7.50
Reviews	0.97	1.96	0.55	0.62	1.14	5.48
Wiki	1.25	3.65	1.61	3.54	0.83	6.26
Mean (full)	1.23	3.13	1.52	1.73	1.07	8.15

Table 10: Per-domain selected flip rate (%) at $k =$ full, 15-cell means. Domain means may differ slightly from the overall mean due to variation in the number of valid-human pairs per domain. GPT-2 shows elevated flip rates on news, Reddit, and wiki (3.1–4.9%) relative to books and reviews (1.9–2.0%), which suggests its detection signal is stronger in formal or community text domains. Cohere is uniformly high across all six domains (5.5–10.1%), because its probe operates near the decision boundary throughout. Mistral shows a notable wiki spike (3.54%) while reviews are weak (0.62%).

Layer	GPT-4	GPT-2	MPT	Mistral	LLaMA	Cohere
1–10	36	41	**44**	24	30	40
11	6	**11**	6	5	2	3
12	**18**	8	2	**16**	**16**	**19**
% in L12	30.0	13.3	3.8	35.6	33.3	30.6

Table 11: Stable-neuron counts by layer group across six generators. Bold indicates the peak layer group for each generator. Instruction-tuned generators (GPT-4, Mistral, LLaMA, Cohere) concentrate 30–36% of stable neurons in layer 12; both pure-base generators (GPT-2, MPT) have $\leq$14% in layer 12.

Pair	Jaccard	$\times$ chance
Cohere – Mistral	0.244	86$\times$
LLaMA – Mistral	0.163	64$\times$
GPT-4 – Mistral	0.141	51$\times$
GPT-4 – LLaMA	0.113	39$\times$
MPT – GPT-2	**0.120**	**40**$\times$
Cohere – GPT-4	0.109	33$\times$
Cohere – LLaMA	0.078	27$\times$
GPT-2 – Mistral	0.050	18$\times$
Cohere – GPT-2	0.034	10$\times$
GPT-2 – LLaMA	0.019	7$\times$
MPT – Mistral	0.010	4$\times$
MPT – LLaMA	0.010	4$\times$
GPT-2 – GPT-4	0.000	0$\times$
MPT – GPT-4	0.000	0$\times$
MPT – Cohere	0.000	0$\times$
Mean	0.073	$\sim$24$\times$

Table 12: Pairwise Jaccard similarity between stable sets (15 generator pairs). $\times$ chance is observed Jaccard divided by the analytic random-null Jaccard for two sets of the same sizes drawn from 9,216 neurons ($\approx$0.003 per pair). The mpt–gpt2 pair (bold) is the highest base$\times$base overlap; all base$\times$instruction-tuned pairs cluster at the bottom.

k	GPT-4	GPT-2	MPT	Mistral	LLaMA	Cohere
Forward direction (AI donor → human target)						
1	0.02	0.05	0.04	0.06	0.07	0.13
5	0.09	0.17	0.14	0.20	0.14	0.58
10	0.15	0.29	0.25	0.34	0.27	0.99
20	0.35	0.58	0.47	0.56	0.49	2.02
50	0.80	1.71	1.13	1.48	0.86	5.43
full	$1.23_{\pm 0.33}$	$3.13_{\pm 0.54}$	$1.52_{\pm 0.22}$	$1.73_{\pm 0.27}$	$1.07_{\pm 0.37}$	$8.15_{\pm 1.27}$
Reverse direction (human donor → AI target)						
full	$0.65_{\pm 0.21}$	$2.41_{\pm 0.62}$	$1.36_{\pm 0.40}$	$1.35_{\pm 0.27}$	$0.92_{\pm 0.21}$	$5.74_{\pm 0.87}$

Table 13: Flip rate (%) for both patching directions, 15-cell means (5 folds $\times$ 3 seeds). k = full equals each cell's actual $|\mathcal{S}|$ (53–92 neurons); std shown only for k = full. **Forward direction** (AI donor → human target) provides the k-sweep and is monotonically increasing in k for every generator. **Reverse direction** (human donor → AI target) was evaluated only at k = full; rates are consistently 7–12$\times$ above the random-k baseline and lower than the forward direction, consistent with the partial redundancy interpretation in §5.1.2.